

# Scalable, open-access and multidisciplinary data integration pipeline for climate-sensitive diseases

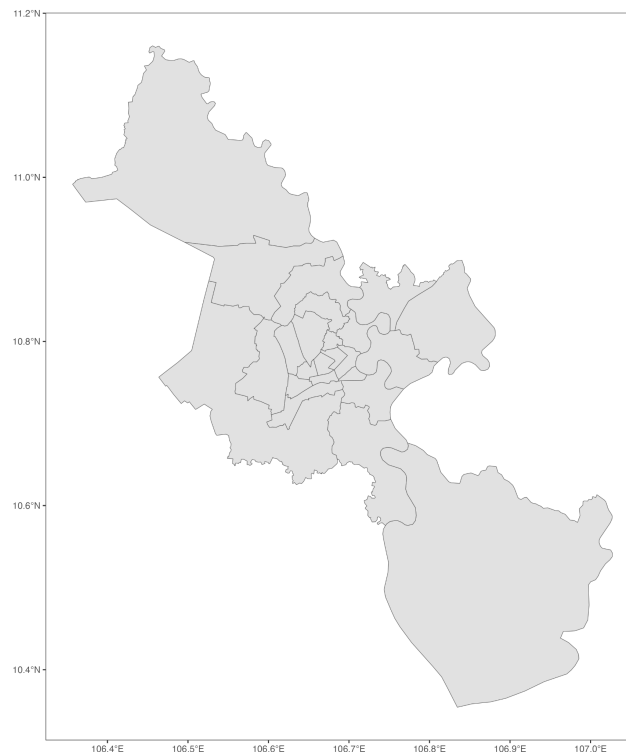
## SUPPLEMENTARY MATERIAL

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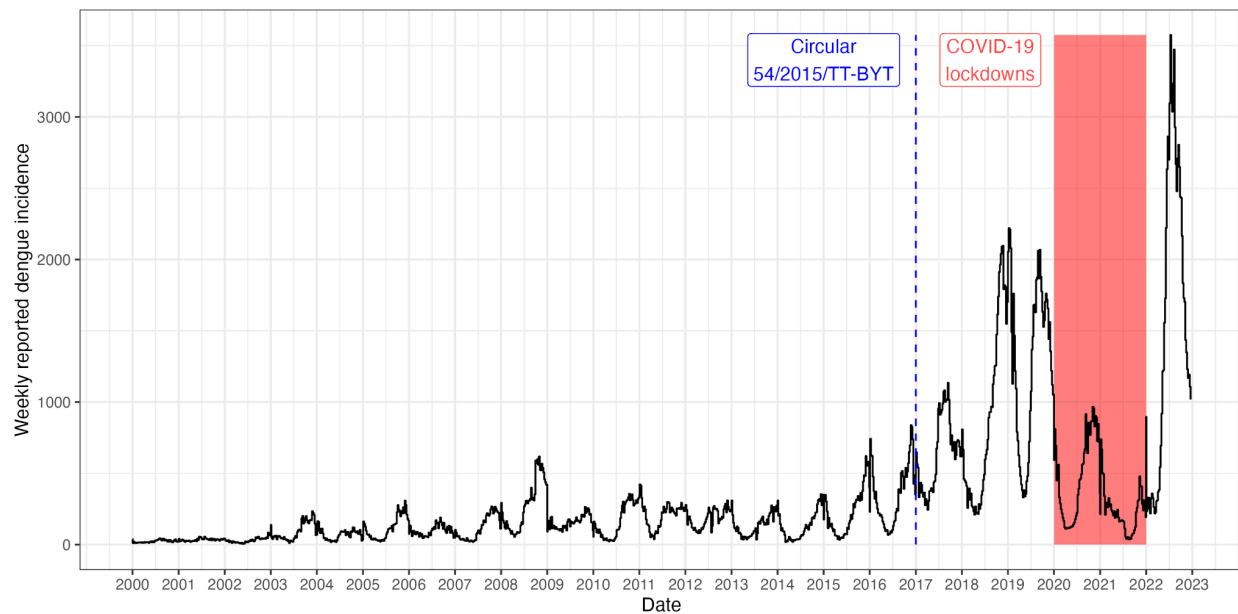
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**MAIN ARTICLE:** <https://wellcomeopenresearch.org/articles/10-467>

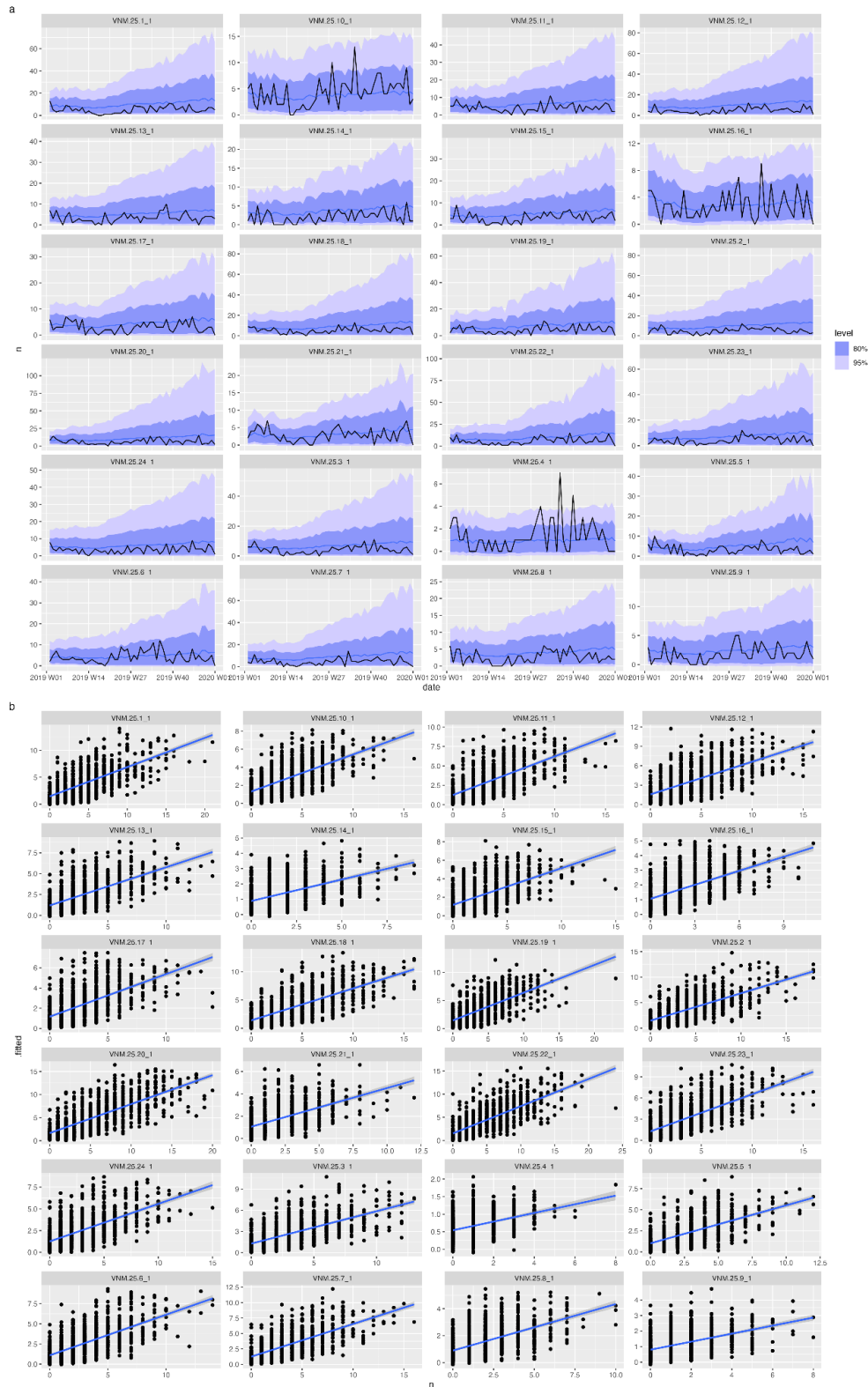
**Figure S1.** Map of administrative boundaries of the 24 districts (Global Administrative Level 2) of the Ho Chi Minh City (HCMC) province (Global Administrative Level 1), Vietnam, in 2020.



**Figure S2.** Reported dengue incidence in HCMC, Vietnam, from 2000 to 2022. From 2000 to 2016, only in-patient admissions were recorded, with data collected and validated manually by HCDC. At the start of 2017, all hospitals and clinics in Vietnam began using a new infectious disease incidence reporting portal (47) to report case admissions for both in-patient and out-patient. This change resulted in substantial increases in the observed dengue incidence. This rise in reported cases is not necessarily indicative of an unusual outbreak but instead reflects the ability of smaller hospitals and clinics throughout the city to report data to the national system. HCDC collaborates with this system by filtering and downloading incidence reports for patients whose residential addresses are in HCMC. They then manually validate these addresses to confirm that said patients reside in HCMC.



**Figure S3.** Predicted dengue incidence from SARIMAX models trained using synthetic case incidence data and ERA5 data zonally aggregated from 2001-2019. Figure S3(a) shows the forecast timeseries; Figure S3(b) shows the  $R^2$  values for administrative level 2 regions in Ho Chi Minh City.



## Supplementary tables

**Table S1.** Resampling methods and aggregation schemes for each dataset and variable, used to achieve a common resolution for dengue modelling. Derived variables are calculated after resampling, but before spatial aggregation. All datasets are licensed under CC-BY-4.0 (Creative Commons Attribution 4.0) unless otherwise stated.

| Dataset Source                     | Variable                                                    | Resampling method (to ~1km at equator, 30-arc seconds) | Spatial aggregation                                                                   | Temporal aggregation | Source                                                                                      |
|------------------------------------|-------------------------------------------------------------|--------------------------------------------------------|---------------------------------------------------------------------------------------|----------------------|---------------------------------------------------------------------------------------------|
| <b>Meteorological</b>              |                                                             |                                                        |                                                                                       |                      |                                                                                             |
| ERA5 atmospheric reanalysis (era5) | Mean/max/min 2m temperature (Kelvin)<br>t2m, mn2t24, mx2t24 | Bilinear interpolation (24)                            | Population and area weighted mean of daily mean, mean of daily max, mean of daily min | Weekly mean          | (42)                                                                                        |
|                                    | Total daily precipitation (mm)<br>tp                        | Distance-weighted average remapping (25)               | Population and area weighted sum                                                      | Weekly sum           |                                                                                             |
|                                    | Mean/max/min relative humidity (%)<br>r, mxr24, mnr24       | Bilinear interpolation (24)                            | Population and area weighted mean of daily mean, mean of daily max, mean of daily min | Weekly mean          | Computed using 2m temperature, 2 metre dew point temperature and surface pressure from (43) |
|                                    | Mean/max/min specific humidity (g/kg)<br>q, mxq24, mnq24    |                                                        | Population and area weighted mean of daily mean, mean of daily max, mean of daily min | Weekly mean          |                                                                                             |
|                                    | Hydrological                                                | Distance-weight                                        | Population                                                                            | Weekly sum           | Computed using the                                                                          |

|  |                                                                                                                                                     |                                   |                                        |            |                                                                                                                                 |
|--|-----------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------|----------------------------------------|------------|---------------------------------------------------------------------------------------------------------------------------------|
|  | balance<br>(m)<br><br>hb                                                                                                                            | hted average<br>remapping<br>(25) | and area<br>weighted sum               |            | sum of daily<br>Evaporation plus<br>total precipitation<br>from (43)                                                            |
|  | Standardised<br>Precipitation<br>Index (units<br>of standard<br>precipitation)<br><br>measured in<br>time window<br>(rolling<br>average)<br><br>spi |                                   | Population<br>and area<br>weighted sum | Weekly sum | Computed using<br>daily total<br>precipitation from<br>(43) and following<br>(44)                                               |
|  | Standardised<br>Precipitation-<br>Evaporation<br>Index (units<br>of standard<br>precipitation)<br><br>spei                                          |                                   | Population<br>and area<br>weighted sum | Weekly sum | Computed using<br>daily total<br>precipitation and<br>potential<br>evaporation from<br>(43) and following<br>(45, 46)           |
|  | Corrected<br>total daily<br>precipitation<br>(m)<br><br>tp_bc                                                                                       |                                   | Population<br>and area<br>weighted sum | Weekly sum | Same as <i>Total daily<br/>precipitation</i> , but<br>after applying a<br>bias-correction<br>technique (See<br>Methods section) |
|  | Standardised<br>Precipitation<br>Index using<br>corrected<br>total daily<br>precipitation<br>(units of<br>standard<br>precipitation)<br><br>spi_bc  |                                   | Population<br>and area<br>weighted sum | Weekly sum | Same as<br><i>Standardised<br/>Precipitation Index</i>                                                                          |
|  | Standardised<br>Precipitation-<br>Evaporation<br>Index using                                                                                        |                                   | Population<br>and area<br>weighted sum | Weekly sum | Same as<br><i>Standardised<br/>Precipitation-Evapo<br/>transpiration Index</i>                                                  |

|                                                                                          |                                                                                      |                                           |                                         |                                         |                                        |
|------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|-------------------------------------------|-----------------------------------------|-----------------------------------------|----------------------------------------|
|                                                                                          | corrected total daily precipitation (units of standard precipitation)<br><br>spei_bc |                                           |                                         |                                         |                                        |
|                                                                                          | Hydrological balance using corrected total daily precipitation (m)<br><br>hb_bc      |                                           |                                         |                                         |                                        |
| ECMWF extended weather forecast data 2 weeks ahead (47)                                  | Same variables as ERA5, except hydrological balance.                                 | See above for details for each variable   | See above for details for each variable | See above for details for each variable |                                        |
| <b>Socio-demographic</b>                                                                 |                                                                                      |                                           |                                         |                                         |                                        |
| WorldPop 2000-2030<br><br>2021-2030 are predicted estimates                              | Population counts<br><br>pop                                                         | NA (already at desired target resolution) | Area weighted sum                       | Yearly                                  | 2000-2020: (48)<br><br>2021-2030: (49) |
| Meta<br><br>Creative Commons Attribution-Non Commercial 4.0 International (CC BY-NC 4.0) | Relative wealth index<br><br>rwi                                                     | Available at 2.4km resolution             | Population weighted mean                | Static (last uploaded April 8, 2021)    | (50)                                   |

**Table S2.** Formulae for spatial aggregation used by *exactextract* (51). Here  $x_i$  is the weather raster at pixel  $i$  resampled to be of the same resolution as the population raster,  $w_i$  is the population raster, and  $c_i$  is the coverage weight, which is the fraction of the polygon covered by both weather and population rasters multiplied by the spherical area approximation of pixel  $i$  in  $\text{km}^2$ .

| Aggregation type  | Formula                                               |
|-------------------|-------------------------------------------------------|
| weighted_mean     | $\frac{\sum x_i c_i w_i}{\sum c_i w_i}$               |
| area_weighted_sum | $\frac{\sum x_i c_i w_i}{\sum c_i}$                   |
| weighted_variance | $\frac{\sum c_i w_i (x_i - \bar{x})^2}{\sum c_i w_i}$ |
| mean              | $\frac{\sum x_i c_i}{\sum c_i}$                       |
| sum               | $\sum x_i c_i$                                        |
| count             | $\sum c_i$                                            |
| weighted_sum      | $\sum x_i c_i w_i$                                    |

**Table S3.** Automated validation checks by variable. We conduct automated range checks for a subset of variables in addition to ensuring that no variable has an NA value.

| Metric                   | Range                                                                                                                    |
|--------------------------|--------------------------------------------------------------------------------------------------------------------------|
| era5.2m_temperature      | 175 — 325 K (-98.15 — 61.85 °C)<br>These are slightly wider than the actual lowest and highest air temperatures recorded |
| era5.total_precipitation | 0 — (no upper bound defined)                                                                                             |
| era5.wind_speed          | 0 — 110 m/s                                                                                                              |
| era5.relative_humidity   | 0 — 100 (%)                                                                                                              |
| era5.specific_humidity   | 0 — (no upper bound defined)                                                                                             |